BAR ORYAN, Ph.D.

Green Postdoctoral Fellow, Scripps Institution of Oceanography

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Education and Work Experience

Green Postdoctoral Fellow: *Jan 2024 - Current*
Institute of Geophysics and Planetary Physics, Scripps Institution of Oceanography, La Jolla, California, USA.
Advisor: Prof. Alice Gabriel

Postdoctoral Research Scientist: *Mar 2022– Dec 2023*
École Normale Supérieure, Paris, France.
Advisor: Dr. Jean-Arthur Olive
Prof. Romain Jolivet

Doctor of Philosophy in Geophysics: *Sep 2016 – Feb 2022*
Lamont-Doherty Earth Observatory, Columbia University, New York, USA.
Advisors: Prof. Roger W. Buck.
Prof. Michael Steckler.

Master of Science in Geophysics: *Jul 2014 - Jul 2016*
Tel Aviv University, Tel Aviv, Israel.
Advisor: Prof. Zvi Ben-Avraham.

Bachelor of Science in Physics and Geosciences: *Oct 2010- Jun 2014*
Tel Aviv University, Tel Aviv, Israel.

Publications

- **Oryan, B.**, Gailleton, B., Olive, J.-A., Malatesta, L., & Jolivet, R. (2025). Inferring Long-Term Tectonic Uplift Patterns from Bayesian Inversion of Fluvially-Incised Landscapes, *Journal of Geophysical Research: Solid Earth* (https://doi.org/10.1029/2024JB030819)
- Chong, J.-H., **Oryan, B.**, Shen L., Steckler, M. S., & Lindsey, E. O. (2025). Interseismic uplift of anticlines above the Rakhine-Bangladesh Megathrust from ALOS-2 InSAR, *Journal of Geophysical Research: Solid Earth* (http://dx.doi.org/10.1029/2024JB030003).
- Kutschera, F., Jia, Z., **Oryan, B.**, Wong, J. W. C., Fan, W., & Gabriel, A. A. (2024). The Multi-Segment Complexity of the 2024 M_w 7.5 Noto Peninsula Earthquake Governs Tsunami Generation, *Geophysical Research Letters* 51.21 (2024): e2024GL109790.
- **Oryan, B.,** Olive, J. A., Jolivet, R., Malatesta, L. C., Gailleton, B., & Bruhat, L. (2024). Megathrust locking encoded in subduction landscapes. *Science Advances*, *10*(17), eadl4286.

- Steckler, M. S., Jaman, M. H., Grall, C. J., Goodbred, S. L., Wilson, C. A., & **Oryan, B.** (2024). Contribution of campaign GNSS toward parsing subsidence rates by time and depth in coastal Bangladesh. *Frontiers in Earth Science*, *12*, 1354686.
- **Oryan, B.,** Betka, P. M., Steckler, M. S., Nooner, S. L., Lindsey, E. O., Mondal, D., ... & Than, O. (2023). New GNSS and geological data from the Indo-Burman subduction zone indicate active convergence on both a locked megathrust and the Kabaw Fault. *Journal of Geophysical Research: Solid Earth*, *128*(4), e2022JB025550.
- Steckler, M. S., **Oryan, B.,** Wilson, C. A., Grall, C., Nooner, S. L., Mondal, D. R., ... & Goodbred, S. L. (2022). Synthesis of the distribution of subsidence of the lower Ganges-Brahmaputra Delta, Bangladesh. *Earth-Science Reviews*, *224*, 103887.
- **Oryan, B.,** & Savage, H., (2021) Regional heat flow analysis reveals frictionally weak Dead Sea fault. *Geochemistry, Geophysics, Geosystems*, *22*(12), e2021GC010115.
- **Oryan, B**., & Buck, W. R. (2020). Larger tsunamis from megathrust earthquakes where slab dip is reduced. *Nature Geoscience*, 1-6.
- **Oryan, B**., Villinger, H., Lazar, M., Schwab, M. J., Neugebauer, I., & Ben-Avraham, Z. (2019). Heat flow in the Dead Sea from the ICDP boreholes and its implication for the structure of the basin. *Quaternary Science Reviews*, 210, 103-112.
- Malinverno, A., Cook, A. E., Daigle, H., & **Oryan, B**. (2018). Glacial cycles influence marine methane hydrate formation. *Geophysical Research Letters*, 45(2), 724-732.

Under review:

- **Oryan, B.** et al. (50 coauthors, under review in Seismica), The Coupling Cloud: A community database of megathrust kinematic coupling models (https://doi.org/10.31223/X5WJ2R)
- **Oryan, B.** & Gabriel AA (under review in *Nature*). Do coupled megathrusts rupture? (https://doi.org/10.31223/X5HB3N).
- Rao M. P., Pacheco-Solna A., Li R., **Oryan, B**. et al. (under review in *Science Advances*). Atmospheric aridity decouples carbon assimilation and growth in temperate deciduous oaks.
- Gabriel AA, P. Karki, Y. Megan, **B. Oryan**, et al. (under review in Seismological Research Letters). A High-Performance Computing Volumetric Software to Model Sequences of Earthquakes and Aseismic Slip Across Complex Fault Systems (https://doi.org/10.31223/X5V75G)

Awards & Grants

- SCEC workshop grant ($20,000) *2025*
  Tandem Workshop: Modeling the Earthquake Cycle from Faults to Supercomputers.
  Oryan, B., Gabriel, AA. May D., Yun J., Megen Y.
- CRESCENT Geoscience Professional Development Fellowship.
- Cecil H. and Ida M. Green Foundation for Earth Sciences ($3,000). *2025*
  Tandem Supercomputing Hackathon. **Oryan, B.**, Gabriel AA, May D.
- Top 10% Most-Read Author, *Journal of Geophysical Research: Solid Earth*, 2023. *2025*
- Cecil H. and Ida M. Green Foundation for Earth Sciences ($2,000). *2024*
  Seismic Cycle and Aseismic Slip Supercomputing Hackathon. **Oryan, B.** Gabriel AA, and May D.
- CRESCENT seed grant ($30,000). *2024*

Synergizing megathrust Seismo-Geodetic coupling and slip models using Optimal Transport and Machine Learning Frameworks to mitigate earthquake hazard in Cascadia. **Oryan, B**. and Gabriel, AA.

- IGPP Green Postdoctoral Fellowship ($149,000). *2023*
- NASA Earth Surface and Interior grant ($650,000). *2022*
  GNSS and InSAR Surface Deformation Constraints on a 3-D Geodetic Model of the IndoBurma Subduction Zone. Steckler M., Lindsey E, **Oryan B**., et al.
- AGU 2020 Outstanding Student Presentation Award. *2021*
- Lamont-Doherty Earth Observatory Climate Center ($10,000). *2021*
  Temporal dynamics of tree-growth and photosynthesis and their environmental drivers in the Lamont Sanctuary Forest Preserve. Rao M., Pacheco-Solana B., **Oryan, B**., et al.
- Chevron Student Incentive Fund ($3,200). *2020*
  Developing the LDEO PhenoCam network to track the fate of forest carbon from photosynthesis to growth. **Oryan, B**. and Rao M.
- Chateaubriand Fellowship ($6,000). *2020*
  Finical support to work with Dr. Jean-Arthur Olive at the Laboratoire de Géologie de l'Ecole Normale Supérieure (ENS).
- Stork Fund ($12,500). *2019*
  Dept. of Earth and Environmental Sciences graduate student fieldtrip to Peru. **Oryan, B**. and Myers, E.
- Dean's fellow, Department of Earth and Environmental Sciences, Columbia University. *2016*
- M.Sc. Excellence Scholarship, Tel Aviv University ($1,000) *2015*

Selected Presentations

- AGU 2025 (talk): Dec 2025
  Locked but not Loaded? A Global Analysis of Megathrust Coupling and Earthquake Slip
- University of Nevada, Department of Geological Sciences and Engineering (invited talk) Nov 2025
- SSA 2025 (talk) Apr 2025
  Rock, River, Record: Reading the Geomorphic Record of Seismic Cycles Through Bayesian Inversion of River-incised Landscapes
- Rutgers University, Department of Earth and Planetary Sciences (invited talk) Mar 2025
- Dartmouth College, Department of Earth Science (invited talk) Feb 2025
- University of Oregon, Department of Earth Sciences (invited talk) Jan 2025
- AGU 2024 (invited talk): Dec 2024
  Subduction sculpting when megathrusts sleep: How interseismic deformation encodes a signature of plate locking in forearc landscapes.
- AGU 2024 (talk): Dec 2024
  Inferring Spatially-Variable Rock Uplift from Fluvially-Incised Landscapes: a Bayesian Inversion Framework in χ-space.
- IRN-Andes-FRENSZ seminar (invited talk): Oct 2024
  Megathrust locking encoded in subduction landscapes.
- AGU 2022 (talk): Dec 2022
  Spatial patterns of long-term forearc uplift inferred from river profiles, and their comparison with short-term deformation
- AGU 2022 (talk): Dec 2022

New GNSS and geological data from the Indo-Burman subduction zone indicate active convergence on both a locked megathrust and the Kabaw Fault

- JpGU 2022 (invited talk) — May 2022
- EGU 2022 (talk) — May 2022
- AGU Fall 2021 (invited talk) — Dec 2021
- Caltech Seismo Lab Seminar (invited talk) — Oct 2021

Teaching Experience

- Co-Mentor, Heat advection experiment, École Normale Supérieure. — *Fall 2022*
  Guided students in conducting and analyzing a heat advection experiment.
- Instructor, Field Trip to Peru, Department of Earth and Environmental Sciences, Columbia university. — *Spring-Summer 2019*
  Led and organized a geological field trip to Peru.
- Mentor, Lamont-Doherty Earth Observatory Summer intern program. — *Summer 2019*
  Served as mentor for a group of low-income high school students.
- Teaching assistant, Life Systems, Department of Earth and Environmental Sciences, Columbia University. — *Spring 2019*
- Teaching assistant, Geodynamics, Department of Earth and Environmental Sciences, Columbia University. — *Fall 2018*
- Teaching assistant, Lab in Geosciences, Tel Aviv University. — *Fall 2015*
  Designed and developed a lab experiment emulating the heat flow of the Earth for an undergrad class.
- AP calculus and electromagnetism tutor, undergrad students, Tel Aviv University. — *Fall 2014*

Field Work

- HT-RESIST EM research cruise, New Zealand. — *Winter 2019*
  Deployment and recovery of 120 EM receivers as well as 500 line-km of EM source.
- Borehole temperature profile measurements, Dead Sea, Israel. — *Summer 2015*
  ICDP Dead Sea borehole temperature measurement.
- Thermal conductivity measurements, IODP core repository, Bremen, Germany. — *Summer 2014*
  Thermal conductivity measurements of the ICDP Dead Sea cores using KD2 pro probe and optical apparatus.

Outreach

- Venn plate tectonics happy hour. — *2023-2025*
- Science writer at the Little Big Science NGO. — *2018 - 2025*
- Lamont Doherty research as art committee. — *2018-2019*
- Lamont Doherty open house. — *2016 - 2021*
- Earth-Sun Day at the American Museum of Natural History. — *2017-2019*